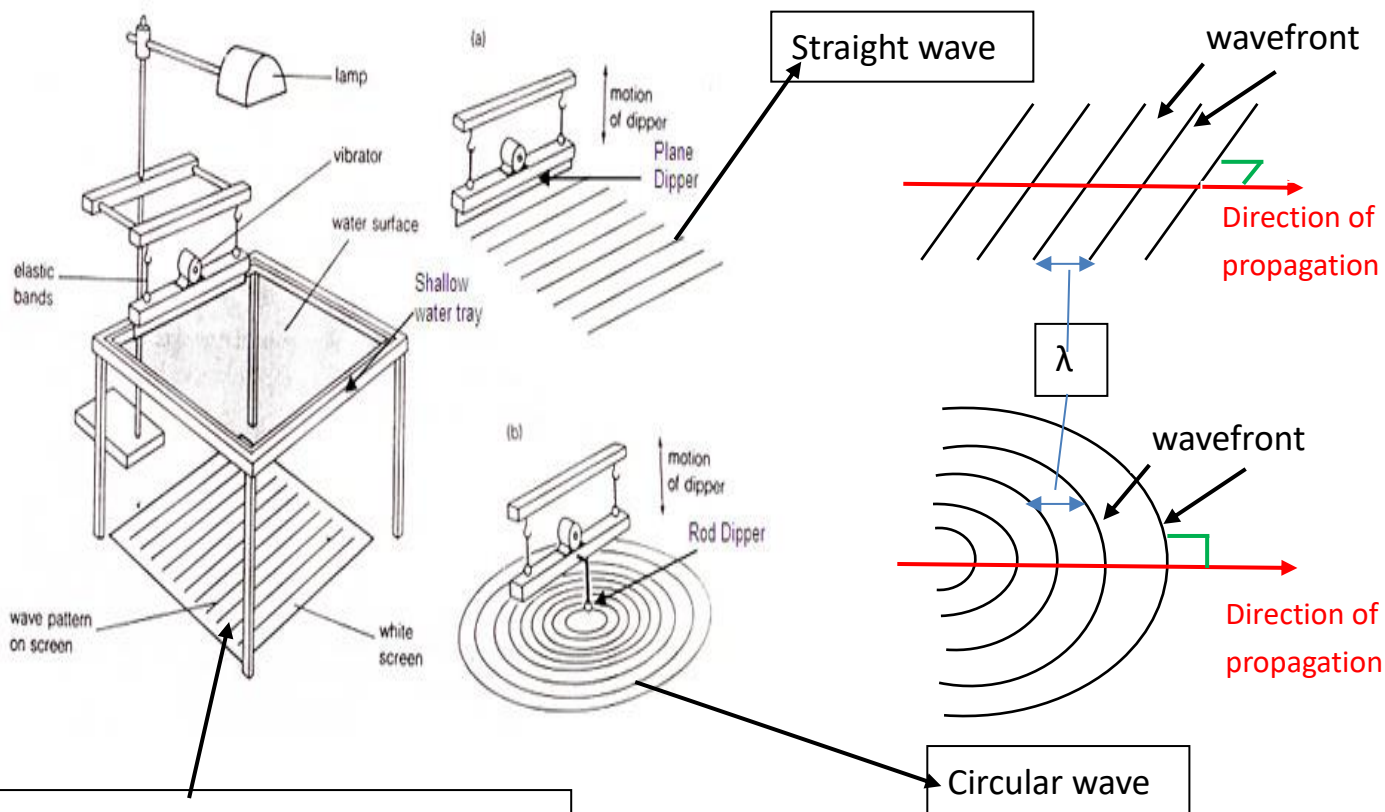


Water wave are transverse waves of which the behaviour can be demonstrated easily by using a **ripple tank** as shown below.

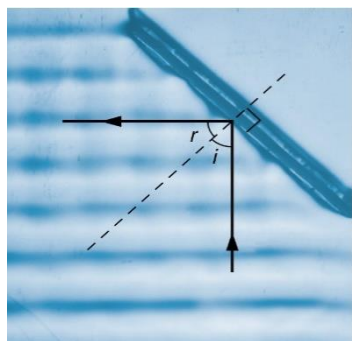


A **bright line** on the screen represents a wave **crest** while a **dark line** represents a wave **trough**.

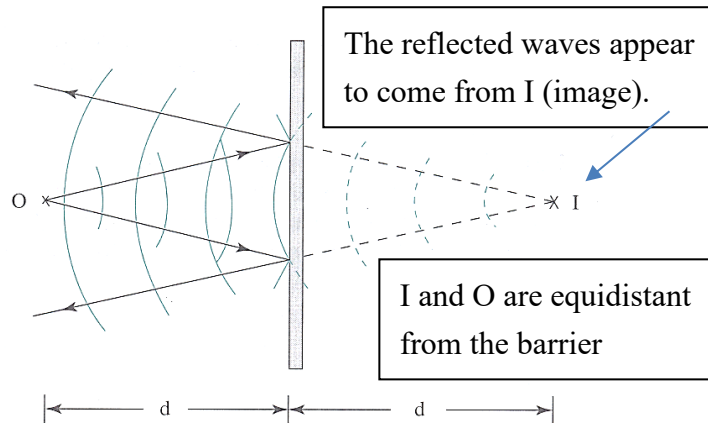
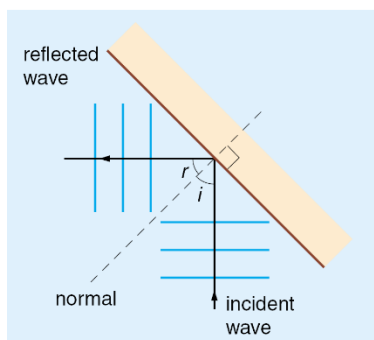
Reflection

Law of reflection:

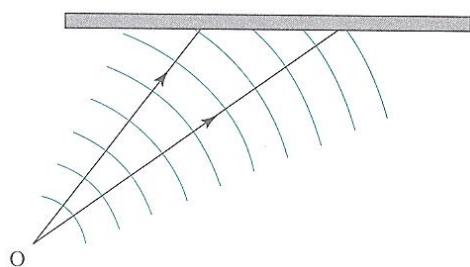
1. Angle of incidence(i) = Angle of reflection(r)
 2. The normal, incident ray and reflected ray all lie on the same plane.
- (Remarks: During reflection, there is no change in speed, wavelength and frequency)



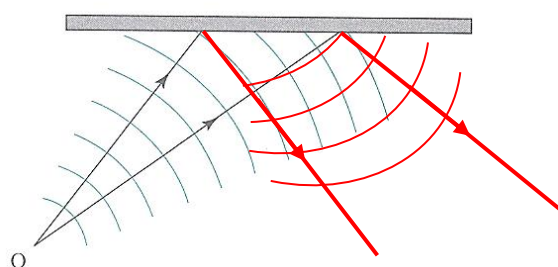
Reflection of a straight wave



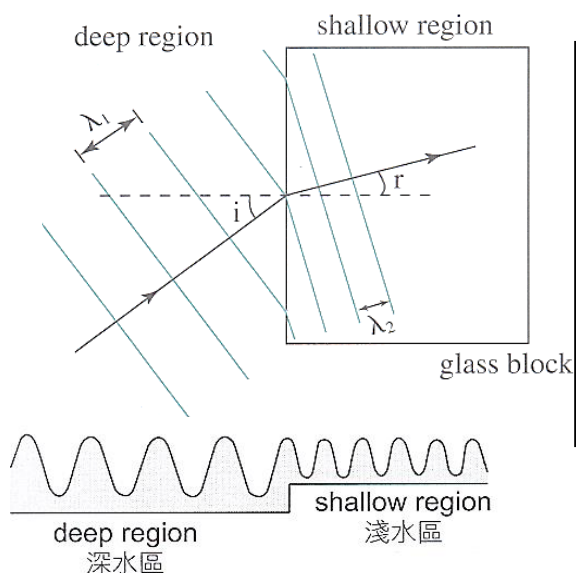
Reflection of a circular wave



1. Sketch the reflected wave.



Refraction (occurs when the **speed of wave is changed** as a wave travels from one medium into another medium)



- When a water wave enters from **deep region** to **shallow region**, its wave speed is reduced.

$\therefore v = f\lambda$ the frequency remains unchanged,

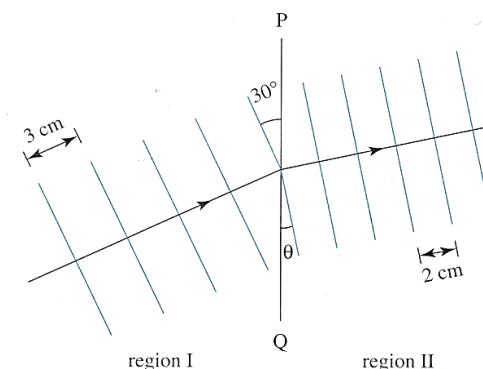
$$v \downarrow \therefore \lambda \downarrow$$

- Relation between the direction, speed and wavelength

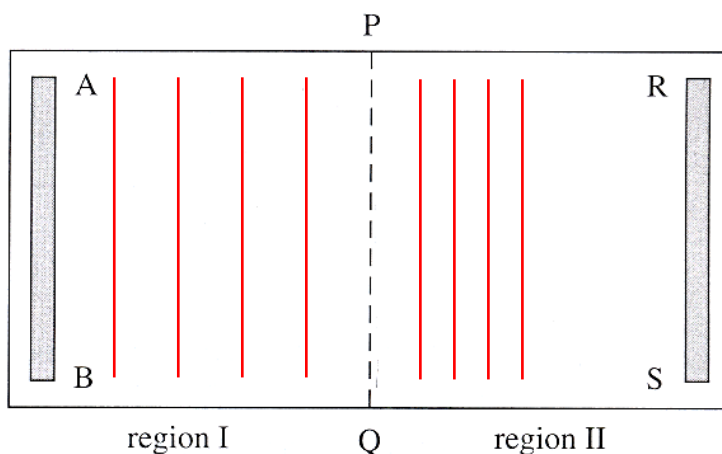
$$\frac{\sin \theta_1}{\sin \theta_2} = \frac{v_1}{v_2} = \frac{\lambda_1}{\lambda_2}$$

2. Which of the regions, I or II is deeper?

Answer : I



3. The straight vibrator is now placed in position AB and parallel with it. Draw the wavefronts in region I (deep region) and region II (shallow region).



警告: 不可翻

4. When a series of water waves travels from a deep region to a shallow region, which of the following wave properties will change?

(1) Wavelength (2) Speed (3) Frequency

- A (1) only
B (2) only
C (1) and (2) only
D (1), (2) and (3)

(1) ✓ See explanation below.
(2) ✓ When wave travels from one medium to another, its speed will change due to refraction. By $v = f\lambda$, frequency is constant, so its wavelength changes.
(3) ✗ The frequency of a wave depends on the frequency of its source only.

C

5. When a plane water wave travels from a shallow region to a deep region of water, the wavelength of the water wave changes. Which of the following statements is/are correct?

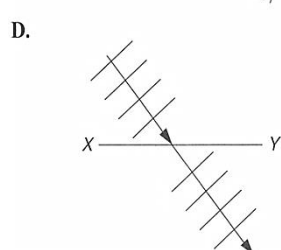
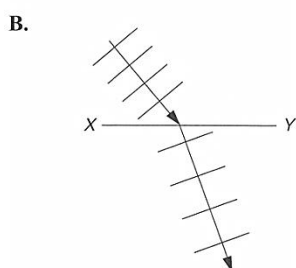
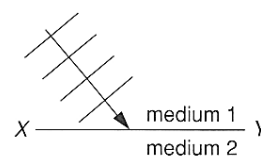
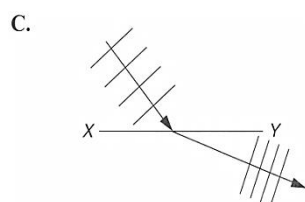
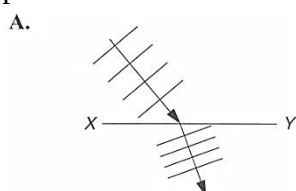
- (1) The wavelength in the shallow region is longer than that in the deep region
(2) The phenomenon is known as refraction
(3) The amplitude of the water wave is reduced in the deep region

- A (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

D
(1) Water waves travel faster in deep region than in shallow region.
(3) Some water waves are reflected.

D

6. XY is a boundary between two media, medium 1 and 2. A series of plane wave travels from medium 1 to medium 2 as shown in the figure. Which of the following figures shows the possible propagation of the plane waves?



A
If the wave speed is reduced after refraction, the wavelength should be smaller and the wave should be refracted towards the normal.

A

7. The figure shows wave crests moving in the direction of the arrow towards the interface PQ between a shallow region and a deep region as shown in the figure. Which of the lines shown may represent one of the wave crests in the deep region?

- A. I
B. II
C. III
D. IV

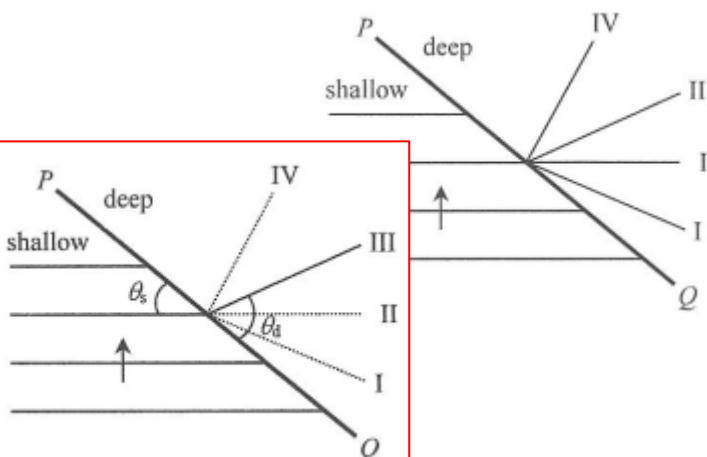
Water wave in deeper region moves with a greater speed.

$$\therefore v_d > v_s \Rightarrow \theta_d > \theta_s$$

$\therefore$ III is the possible wavefront

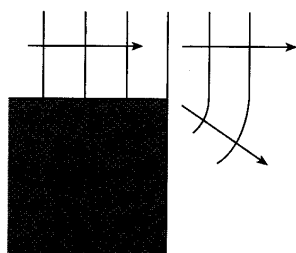
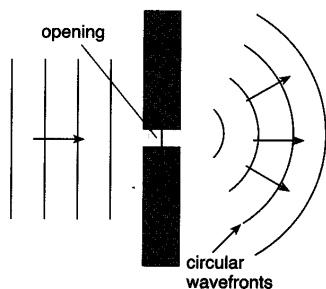
IV is not correct

since the wavefront makes a refracted angle greater than 90° with the boundary, which is impossible.



C

Diffraction



Diffraction is the spreading of a wave when the wave

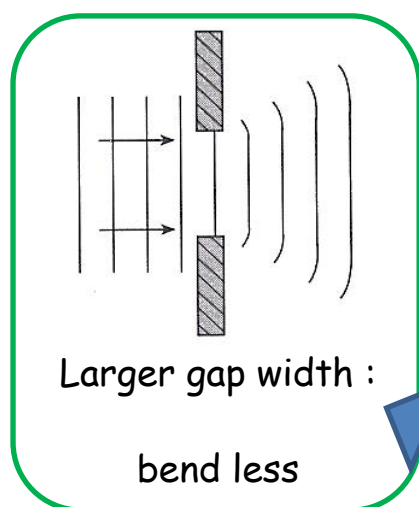
- emerges through a narrow gap in a barrier; and/or
- passes the corner of a barrier

Degree of diffraction increased by

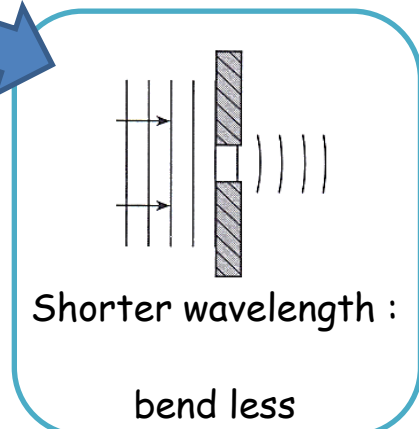
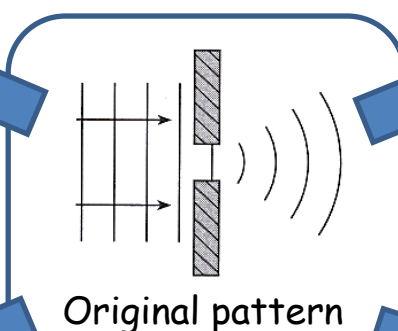
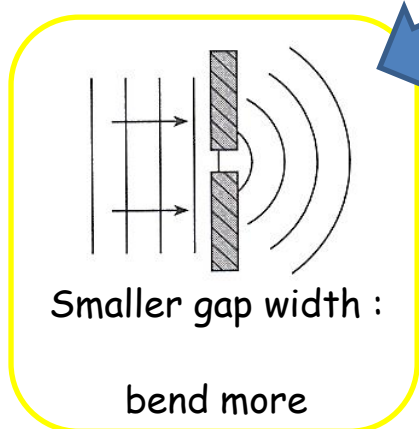
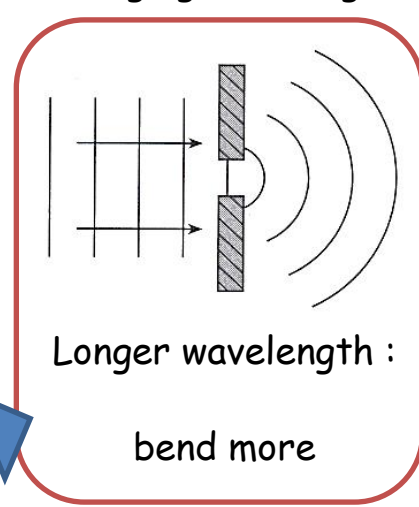
- **larger** wavelength
- **smaller** gap width

(Remarks: During diffraction, there is no change in speed, wavelength and frequency but wave travel and wave pattern are changed)

Changing gap width



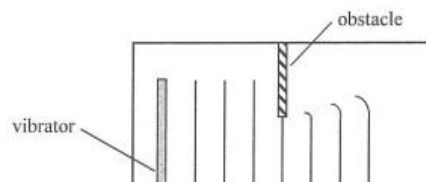
Changing wavelength



8. A student uses the above set-up to study the diffraction of water waves. Which of the following changes can make the diffraction effect more significant ?

- (1) reducing the width of the gap between the obstacles
- (2) increasing the frequency of the vibrator
- (3) adding more water to the ripple tank

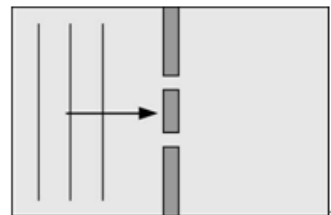
- A. (1) only
B. (2) only
C. (1) & (3) only
D. (2) & (3) only



- | | | |
|---|-----|---|
| ✓ | (1) | A smaller gap can give greater degree of diffraction. |
| ✗ | (2) | By increasing the frequency, the wavelength is decreased. Thus, the degree of diffraction is decreased. |
| ✓ | (3) | By adding more water, depth is increased, thus speed is increased, wavelength is then increased. Therefore, the degree of diffraction is increased. |

9. In a ripple experiment, a series of water waves passes through two gaps as shown. The depth of water in the tank remains unchanged. Which of the following quantities of the wave would remain unchanged after the wave passes through the gaps?

- A Wavelength
B Period
C Speed
D All of the above



D

10. Plane water waves travel from a deep region to shallow region of water through a narrow gap as shown above. Which of the following properties of the waves remains unchanged?

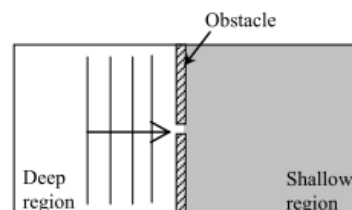
- A. Direction of travel
B. speed
C. wavelength
D. frequency

The frequency of the water wave depends on the source which produces it. As a result, it remains unchanged throughout.

For Answer A: Diffraction occurs because of the obstacle and alters the direction of travel as shown above.

For Answer B: Refraction occurs because of the change in the depth of water, and the speed of wave decreases when it enters a shallower region.

For Answer D: By $v = f\lambda$, when the speed of the wave decreases, its wavelength also decreases (since the frequency of the wave is constant).



D

11. In which of the following situations **Must** the direction of travel of a wave change?

- (1) when a wave is reflected by a barrier
(2) when a wave enters from one medium to another medium
(3) when a wave travels through a gap smaller than its wavelength

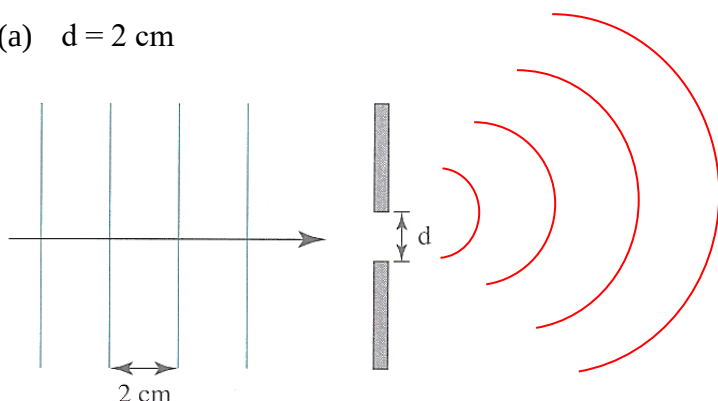
- A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

- (1) ✓ Let θ be the angle between the direction of travel of the incident wave and the normal of the barrier. Note that $0^\circ \leq \theta < 90^\circ$.
By $i = r$, the angle between the direction of travel of the reflected wave and the normal of the barrier is θ . Hence, the wave bends by 2θ . Since $0^\circ \leq 2\theta < 180^\circ$, the direction of travel of the wave must change.
(2) ✗ The direction of travel of the wave remains unchanged if the direction of travel of the incident wave is perpendicular to the interface between the two media.
(3) ✓ The wave bends around the edges of the gap. Therefore, the direction of travel of the wave must change.

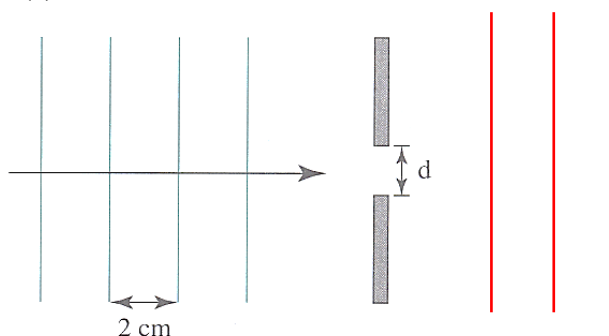
B

12. Draw the diffracted waves if

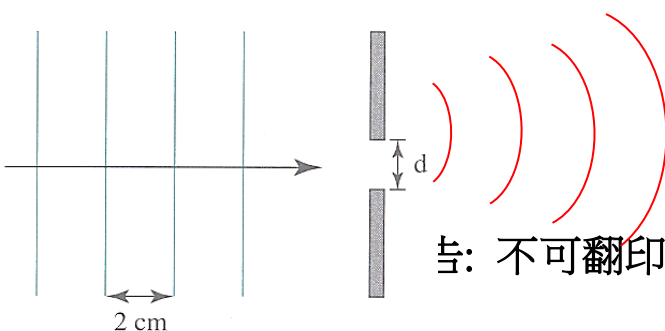
(a) $d = 2 \text{ cm}$



(c) $d = 10 \text{ cm}$



(b) $d = 3 \text{ cm}$



Interference

Principle of superposition:

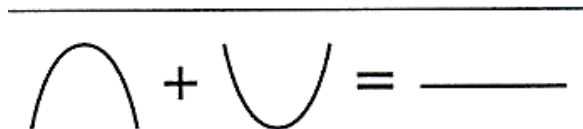
1. two pulses add together to form a resultant pulse
2. displacements of a point on resultant pulse = sum of displacements of individual pulses at that point



trough + trough
= bigger trough



crest + crest
= bigger crest



crest + trough
= no wave

Constructive interference
(Waves arrive in phase)

Destructive interference
(Waves arrive in antiphase)

- Interference occurs when waves from 2 **coherent sources** meet each other.
- Wave from 2 coherent sources:
 - (i) Same frequency
 - (ii) Constant phase relationship
- **Path difference** is the difference between the distances from the two sources to the point of observation. (expressed in λ)
- **Constructive interference** occurs
 - (i) At position where two waves arrive **in phase**. Particle at those positions vibrate with a **larger amplitude**.
 - (ii) At position where the path difference Δ is a **whole number of wavelength**. ($= 0\lambda, 1\lambda, 2\lambda, 3\lambda, \dots, n\lambda$) if two sources **vibrate in phase**.
- **Destructive interference** occurs
 - (i) At position where two waves arrive in **antiphase**. Particles at those position **remain stationary**.
 - (ii) At position where the path difference Δ is

$$\frac{1}{2}\lambda, \frac{1}{2}\lambda, \frac{3}{2}\lambda, \frac{5}{2}\lambda, \dots, (n + \frac{1}{2})\lambda$$
 if two sources **vibrate in phase**.

13. Are waves arriving at P and Q in phase or in antiphase?

P: in phase

Q: antiphase

14. What kind of interference occurs at P and Q respectively?

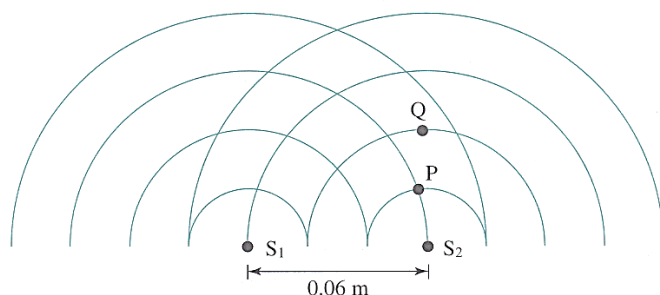
P: constructive interference

Q: destructive interference

15. What is the path difference at P and Q respectively?

P: 2λ

Q: 1.5λ



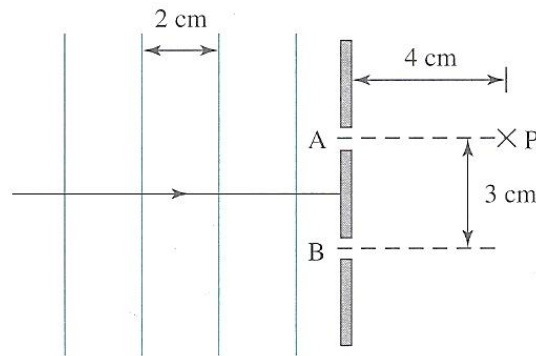
16. What phenomenon is observed at

- (i) A and B? **diffraction**
(ii) P? **interference**

17. What kind of interference occurs at P?

Path difference = 1 cm = 0.5 λ

Destructive interference

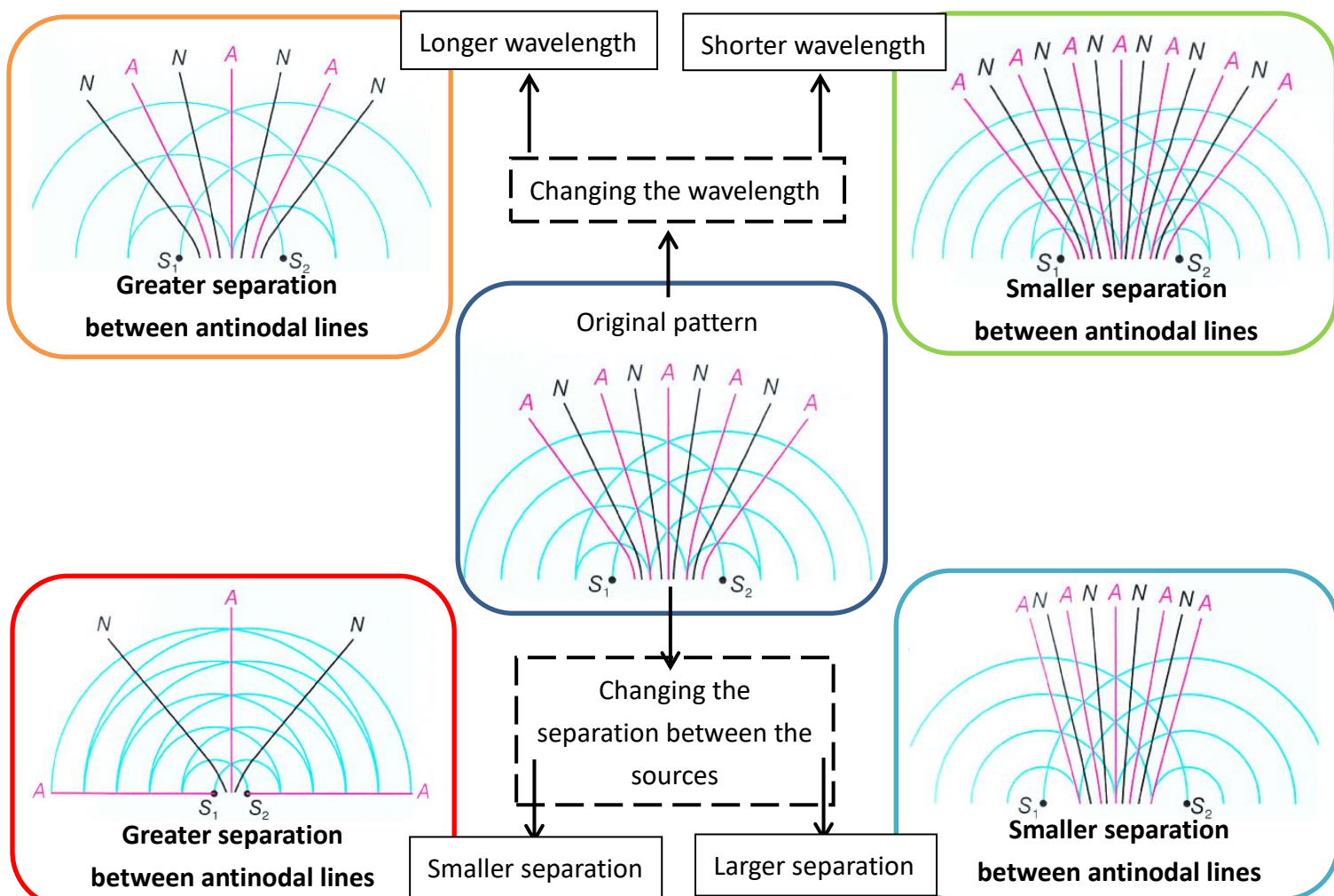
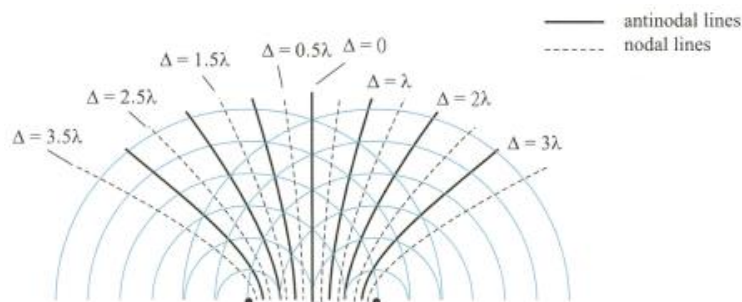


- Points on an **antinodal line** are oscillating with a large amplitude ($2A$). They are in the state of **constructive interference**.

- Points on a **nodal line** have no oscillation, that is, amplitude is zero. They are in the state of **destructive interference**.

Remarks:

- Energy at the points of destructive interference is **redistributed** to points of constructive interference.
- Displacement of particles along a **nodal line** is always **zero**.



警告: 不可翻印, 販賣, 違者必究

18. In the following figure, a plane wave travels towards two narrow gaps and circular waves come out from the gaps. P is 15 cm away from S_1 and 20 cm away from S_2 , while Q is 14 cm away from S_1 and 12 cm away from S_2 . The wavelengths of the waves are 2 cm. If the frequency of the vibrator is doubled, what kind of interference will occur at P and Q ?



A

- | | P | Q |
|---|--------------|--------------|
| A | Constructive | Constructive |
| B | Constructive | Destructive |
| C | Destructive | Constructive |
| D | Destructive | Destructive |

19. S_1 and S_2 are two coherent wave sources. When the path difference from these two sources to a point P is 0.6 m, constructive interference occurs at point P . When the path difference from these two sources to a point Q is 1.05 m, destructive interference occurs at point Q . Which of the following is the possible wavelength of the waves?

	Path difference	
	Constructive interference $0.6 = m\lambda$	Destructive interference $1.05 = (m + \frac{1}{2})\lambda$
$m = 1$	$\lambda = 0.6$	$\lambda = 0.7$
$m = 2$	$\lambda = 0.3$	$\lambda = 0.42$
$m = 3$	$\lambda = 0.2$	$\lambda = 0.3$

C

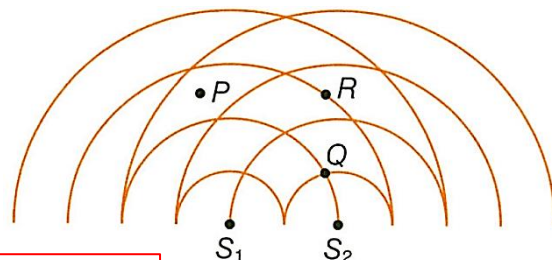
- A 0.2 m
B 0.25 m
C 0.3 m
D 0.4 m

20. Circular waves of wavelength λ are produced by two coherent sources S_1 and S_2 as shown. Solid lines represent crests.

Which of the following statements are correct?

- (1) A trough is always formed at P .
(2) The wave from S_1 and S_2 are in phase at Q .
(3) The path difference at R is 0.5λ .

- A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. All of the above



C Constructive interference occurs at P . Therefore, P vibrates with a large amplitude.
 $\therefore$ (1) is incorrect.
Constructive interference occurs at Q . Therefore, the waves from the two sources arrive there in phase.
 $\therefore$ (2) is correct.
Path difference at $R = 3\lambda - 2\frac{1}{2}\lambda = \frac{1}{2}\lambda$
 $\therefore$ (3) is correct.

MISCONCEPTION
Some students wrongly think that a trough is always formed when a trough meets a trough or when destructive interference occurs.

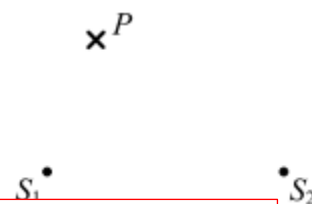
C

21. When two waves coming from two point sources S_1 and S_2 meet at P , they undergo constructive interference. Which of the following statements is/are correct?

- (1) The waves meet at P are in phase.
(2) The path difference at P is equal to the whole number of wavelengths.
(3) The two sources of waves, S_1 and S_2 , must be in phase.

- A. (1) only
B. (3) only

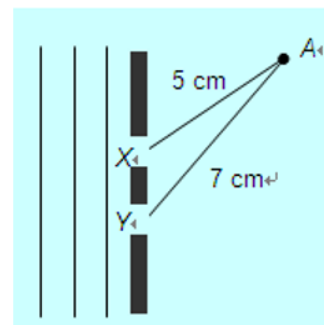
- C. (1) and (2) only
D. (2) and (3) only
- (1) is correct, for constructive interference to occur, the two waves must be in-phase at P to superpose each other and have constructive interference. (2) and (3) are incorrect, if the two sources are anti-phase, the path difference needed for constructive interference is $(m + \frac{1}{2})\lambda$, where m is a whole number.



A

22. Straight water waves are approaching two gaps. The frequency of the waves is 10 Hz and the wave speed is 0.2 m s^{-1} . Which of the following is/are correct?

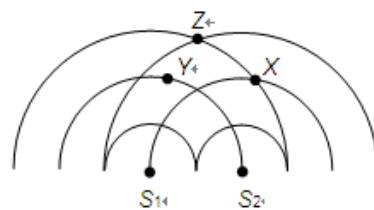
- (1) Destructive interference occurs at A.
 - (2) The waves after passing through the gaps are incoherent.
 - (3) Alternate nodal and antinodal lines are found.
- A (1) only
B (3) only
C (1) and (3) only
D (1), (2) and (3)



B

23. Dippers, S_1 and S_2 , produce circular wavefronts in a ripple tank. Which of the following statements is/are correct?

- (1) Constructive interference occurs at X, Y and Z.
 - (2) Amplitude of the wave at Y is always zero.
 - (3) More nodal and antinodal lines can be observed when the frequency of the dippers increases.
- A (1) only
B (3) only
C (2) and (3) only
D (1), (2) and (3)

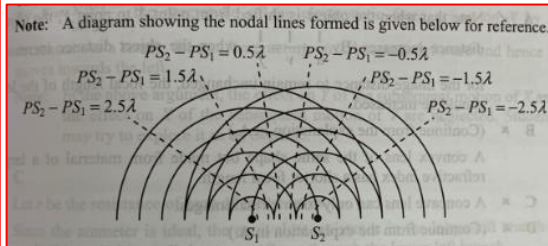


C

24. The figure shows the circular water waves generated by two dippers S_1 and S_2 vibrating in phase. The lines represent wave crests. What is the number of nodal lines (i.e. minimum amplitude) formed?

- A. 3
B. 4
C. 6
D. 7

Let λ be the wavelength of the water waves.
Note that minimum amplitude indicates the occurrence of destructive interference.
Since S_1 and S_2 are vibrating in phase, at any point P on a nodal line, the path difference $PS_2 - PS_1$ equals $\pm 0.5\lambda, \pm 1.5\lambda, \dots$, etc.
Note that a nodal line corresponds to a possible value of the path difference $PS_2 - PS_1$.
By triangle inequality, $PS_2 + S_1S_2 > PS_1$ and $PS_1 + S_1S_2 > PS_2$
 $PS_2 - PS_1 > -S_1S_2$ and $PS_2 - PS_1 < S_1S_2$
 $PS_2 - PS_1 > -3\lambda$ and $PS_2 - PS_1 < 3\lambda$
Hence, there are 6 possible values of the path difference $PS_2 - PS_1$, namely $-2.5\lambda, -1.5\lambda, -0.5\lambda, 0.5\lambda, 1.5\lambda$ and 2.5λ .
Therefore, the number of nodal lines formed is 6.



C

25. In a ripple tank, circular water waves of wavelengths λ_1 and λ_2 ($\lambda_1 > \lambda_2$) are produced by two vibrators S_1 and S_2 respectively. The figure represents the two circular waves propagating on the water surface at a certain moment.

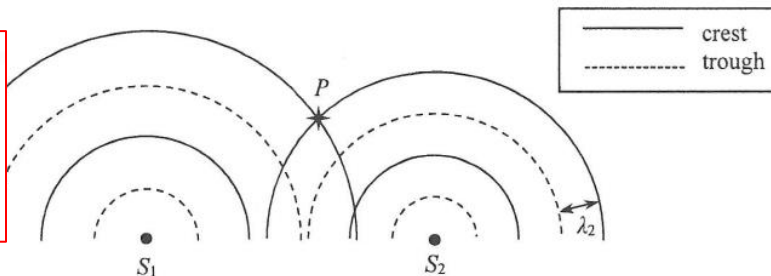
Which of the following statements is correct?

- A. The particle at P is always at crest position.
- B. At P , the two waves always reinforce to give a larger amplitude.
- C. The principle of superposition cannot be applied at P as $\lambda_1 \neq \lambda_2$.
- D. The principle of superposition can be applied at P but the two waves do not always reinforce at that location.

D

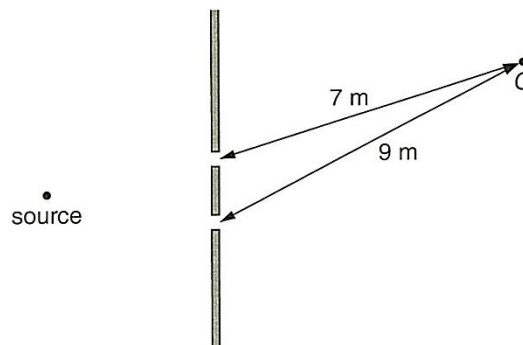
- A * In a travelling water wave, no particle is always at crest position.
B * Since the two waves have different wavelengths, they are not coherent. Hence, no interference pattern is formed. Therefore, at any position, the two waves do not always reinforce to give a larger amplitude.
C * The principle of superposition can be applied at P no matter whether the two waves have equal wavelengths.
D ✓ (See the explanation in B and C.)

警告: 不



26. A source produces water waves of wavelength 2 m and the waves pass through two gaps. O is a small ball floating on the water surface. The distances between O and the gaps are 7 m and 9 m as shown.

- (a) Describe and explain the motion of O .
(b) Describe and explain the motion of O if
(i) the frequency of the source is halved;
(ii) the frequency of the source is doubled.



(a) Path difference at $O = 9 - 7 = 2 \text{ m} = \lambda$

$\therefore$ Constructive interference occurs.

$\therefore$ It vibrates up and down.

(b) (i) By $v = f\lambda$, when the frequency is halved and the wave speed remains unchanged, the wavelength is doubled and equal to 4 m.

$\therefore$ Path difference at $O = 2 \text{ m} = \frac{1}{2}\lambda'$

$\therefore$ Destructive interference occurs. $\bullet$

$\therefore$ It will be at rest.

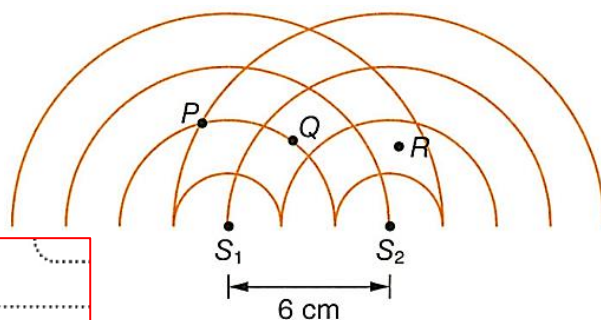
(ii) When the frequency is doubled, the wavelength is halved and equal to 1 m.

$\therefore$ Path difference at $O = 2 \text{ m} = 2\lambda''$

$\therefore$ Constructive interference occurs and it vibrates up and down.

27. Circular waves are produced by two coherent sources S_1 and S_2 in a ripple tank as shown. Solid lines represent crests. The distance between the sources is 6 cm. P , Q and R are three particles on the water surface.

- (i) What is meant by coherent sources?
(ii) What is the wavelength of the waves?
(iii) What are the path differences at P , Q and R ? Hence, describe the motion of the particles.
(iv) Which two particles vibrate in antiphase?
(v) Describe the motion of P and Q if S_2 is removed.



(a) Coherent sources are the sources that have a constant phase difference. $\bullet$

(b) Wavelength = $\frac{6}{3} = 2 \text{ cm}$ $\bullet$

(c) Path difference at $P = 4\lambda - 2\lambda = 2\lambda = 2 \times 2 = 4 \text{ cm}$

Path difference at $Q = 2\frac{1}{2}\lambda - 2\lambda = \frac{1}{2}\lambda = \frac{1}{2} \times 2 = 1 \text{ cm}$

Path difference at $R = 3\frac{1}{2}\lambda - 1\frac{1}{2}\lambda = 2\lambda = 2 \times 2 = 4 \text{ cm}$

$\therefore$ P and R vibrate up and down
and Q is at rest.

(d) P and R $\bullet$

(e) P will vibrate with half of the original amplitude.

Q will vibrate with the same amplitude as P .